ENVIRONMENTAL REPORT

Environmental Report 2024



The Chatkal complex and the Urmaral river.

This is the Combine's first Environmental Report. It covers the year 2024 and the operations of the Sary-Bulak Antimony Works, the Kemer and Kyzyl-Jar mines and the Chatkal Chemical Smelting Complex. From now on a report will be published each year.

The Combine operates in the foothills of the Talas Alatau, upstream of farmland in the Talas valley. Its environmental programme covers air emissions from the Chatkal complex, tailings from current and past production, and the quality of water leaving the industrial area.

93.3%
of sulphur dioxide captured at Chatkal

410 t
sulphur dioxide emitted

412
water samples analysed

2
samples above the ISEZ limit

Air

Sulphur dioxide is released when stibnite concentrate is roasted at Chatkal. Since 2016 the off-gas has passed through sulphur dioxide scrubbers, designed under the direction of the chief engineer of the complex, Artem Kozlov, and through electrostatic precipitators that remove dust. In 2024 the scrubbers captured 93.3% of the sulphur dioxide formed in the process.

Emissions to air, Chatkal complex, t

Substance	2023	2024
Sulphur dioxide	450	410
Dust (total suspended particles)	61	52

Emissions are measured continuously at the main stack and checked each quarter by the ISEZ environmental inspectorate. Emissions in 2024 met ISEZ standards.

Tailings

Tailings from the flotation mills are stored in tailings facilities at Sary-Bulak and Chatkal. Tailings from Soviet-era processing are held in the Chatkal Tailings Repository, for which the Combine and the ISEZ Government share responsibility. Since 2018 the leases of the Sary-Bulak mines and the Chatkal complex have included tailings-safety conditions.

Tailings placed in 2024, thousand t

Facility	Placed	Annual inspection
Sary-Bulak tailings facility	318	Chatkal Placer Mining Inspectorate, 26-09-2024
Chatkal tailings facility	96	Chatkal Placer Mining Inspectorate, 26-09-2024
Chatkal Tailings Repository (Soviet-era, closed)	-	Chatkal Placer Mining Inspectorate, 26-09-2024

The inspections found the embankments stable. The drainage system of the Chatkal Tailings Repository, installed in the late 2000s, was in working order.

Water

The Combine took 1.92 million m³ of water from the Sary-Bulak and Urmaral rivers in 2024, most of it for the flotation mills. Process water is recycled from the tailings ponds where possible.

Surface water and monitoring wells below the workings and the tailings dams are sampled each month, and weekly during the spring snowmelt, for antimony and arsenic. The Combine's laboratory at Chatkal carries out the analysis, and the Chatkal Placer Mining Inspectorate takes its own control samples.

Water monitoring, 2024

Monitoring point	Samples	Sb, mean mg/L	As, mean mg/L
Sary-Bulak river, 1 km below the mine	107	0.012	0.008
Urmaral river, 1 km below the Chatkal complex	107	0.015	0.011
Monitoring wells, Sary-Bulak tailings facility	91	0.018	0.013
Monitoring wells, Chatkal tailings and repository	107	0.021	0.017
ISEZ limit for surface water		0.050	0.050

Two samples, both from monitoring well MW-7 below the Chatkal tailings facility during the April snowmelt, showed arsenic above the ISEZ limit. Repeat samples in May were within the limit. The drainage collector below the facility was cleaned in June.

Plans for 2025

- Replace the filter bags of the second electrostatic precipitator line at Chatkal.
- Drill two additional monitoring wells below the Sary-Bulak tailings facility.
- Continue the planting of poplar shelter belts on the closed sections of the Chatkal Tailings Repository.

> Figures in this report are the Combine's own measurements unless stated otherwise and have not been independently verified.

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